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ORIGINAL INVESTIGATION

Effect of Individual Environmental Heat-Stress Variables on Training and Recovery in Professional Team Sport

Fergus K. O'Connor, Steven E. Stern, Thomas M. Doering, Geoffrey M. Minett, Peter R. Reaburn, Jonathan D. Bartlett, and Vernon G. Coffey

Context: Exercise in hot environments increases body temperature and thermoregulatory strain. However, little is known regarding the magnitude of effect that ambient temperature (T_a), relative humidity (RH), and solar radiation individually have on team-sport athletes. ***Purpose:*** To determine the effect of these individual heat-stress variables on team-sport training performance and recovery. ***Methods:*** Professional Australian Rules Football players (N = 45) undertook 8-wk preseason training producing a total of 579 outdoor field-based observations with T_a, RH, and solar radiation recorded at every training session. External load (distance covered, in m/min; percentage high-speed running [%HSR] >14.4 km/h) was collected via a global positioning system. Internal load (ratings of perceived exertion and heart rate) and recovery (subjective ratings of well-being and heart-rate variability [root mean square of the successive differences]) were monitored throughout the training period. Mixed-effects linear models analyzed relationships between variables using standardized regression coefficients. ***Results:*** Increased solar-radiation exposure was associated with reduced distance covered (−19.7 m/min, $P < .001$), %HSR (−10%, $P < .001$) during training and rMSSD 48 h posttraining (−16.9 ms, $P = .019$). Greater RH was associated with decreased %HSR (−3.4%, $P = .010$) but increased percentage duration >85% HR_{max} (3.9%, $P < .001$), ratings of perceived exertion (1.8 AU, $P < .001$), and self-reported stress 24 h posttraining (−0.11 AU, $P = .002$). In contrast, higher T_a was associated with increased distance covered (19.7 m/min, $P < .001$) and %HSR (3.5%, $P = .005$). ***Conclusions:*** The authors show the importance of considering the individual factors contributing to thermal load in isolation for team-sport athletes and that solar radiation and RH reduce work capacity during team-sport training and have the potential to slow recovery between sessions.

Keywords: exercise training, thermoregulation, physiology

Training and competition in the heat can provide additional physiological demands with the potential to limit the intensity or duration of exercise and increase ratings of perceived exertion (RPE) compared with exercise in thermoneutral conditions.[1] Indeed, there is a large body of evidence examining the effects of hot environments while undertaking steady-state aerobic exercise.[1,2] For example, Galloway and Maughan[3] previously showed that under laboratory conditions of increasing ambient temperature (T_a) with fixed relative humidity (RH), there are significant increases in heart rate (HR), core temperature, skin temperature (T_{sk}), sweat rate, RPE, and impaired exercise capacity. Similarly, when increasing RH with fixed hot T_a, the T_{sk}, sweat rate, and RPE increased and exercise performance decreased.[4] Finally, Otani et al[5] showed that the effects of increased solar radiation (SR) exposure together with constant T_a and RH elevates T_{sk}, reduces the core-to-skin temperature gradient, and subsequently leading to decreased endurance capacity. Accordingly, each heat stress variable contributing to total heat strain may have individual effects with the potential to reduce physical performance in hot conditions.

O'Connor, Doering, Reaburn, Bartlett, and Coffey are with the Faculty of Health Sciences & Medicine, Bond Inst of Health & Sport, and Stern, the Bond Business School, Bond University, Gold Coast, QLD, Australia. O'Connor and Bartlett are also with the Gold Coast Suns FC, Gold Coast, QLD, Australia. Doering is also with the School of Health, Medical and Applied Sciences, Central Queensland University, Rockhampton, QLD, Australia. Minett is with the School of Exercise and Nutrition Sciences and the Inst of Health and Biomedical Innovation, Queensland University of Technology, Brisbane, QLD, Australia. Bartlett is also with the Inst for Health and Sport, Victoria University, Melbourne, VIC, Australia. Coffey (vcoffey@bond.edu.au) is corresponding author.

The vast majority of previous research on thermoregulation during exercise in hot environments has focused on the endurance athlete, and there is a paucity of available data examining the effect of exercise in the heat on the team-sport athlete.[6] Some previous studies have examined physiological changes to heat stress in team-sport players, including characterizing responses during match play at various wet-bulb globe temperatures.[7] Many team sports undertake physical preparation outdoors during summer months at geographical locations characterized by high T_a, RH, and SR.[6] Repeated training in these hot environments may promote long-term adaptations such as improved sweat response and decreased cardiovascular strain with the potential to convey a benefit for team-sport performance.[8] Conversely, the acute thermoregulatory responses to exercise in hot conditions may attenuate physical work capacity and the quality of training sessions, while frequent heat exposure during high-intensity intermittent team-sport training may lead to overreaching and adverse effects on athlete wellness and performance.[9]

Understanding the physiological effects of recurring exposure to heat stress during training or competition is necessary to optimize physical preparation and well-being in team-sport athletes. Moreover, there is currently no data on varying environmental conditions that consider the individual effects of T_a, RH, and SR on external load, internal load, and recovery from intermittent, high-intensity team-sport training activities. Thus, the primary aim of this study was to determine the individual effects of T_a, RH, and SR on professional team-sport training sessions undertaken in hot conditions. Our secondary aim was to establish the individual effect of these heat variables on recovery during the 48-hour period following training bouts. We hypothesized that with increasing

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exposure to each individual heat stress variable, external work completed during training would decrease concomitant with higher internal loads and impaired recovery.

Materials and Methods

Participants

A convenience sample of 45 professional male athletes (mean [SD]; age 22.9 [3.8] y, height 188.4 [8.3] cm, body mass [BM] 86.9 [9.4] kg, maximal aerobic speed 17.6 [0.8] km/h) from a professional football club competing in the Australian Football League participated in this study. Athletes completed a minimum of 5-week preseason training in a hot environment before the data collection period and were deemed to be heat acclimated. Ethical approval was granted by Bond University's human research ethics committee (FO00007).

Experimental Protocol

Internal and external training loads were captured throughout an 8-week training period of the preseason preparation phase spanning January and February of the Australian summer. Upon removal of "stationary skill" sessions that were completed during the experimental period, an average 16 (3) sessions were recorded per player (range: 4–20), resulting in 579 outdoor field-based training observations in the final analysis. Participants wore the same clothing (singlet and shorts) during each training session. Pre and post each training session, BM was recorded with participants wearing the same clothing (running shorts) using standardized weighing scales (Excell Precision, New Taipei, Taiwan) to the nearest 100 g for calculation of BM changes from presession to postsession.

External Training Load

External training loads for each participant were collected and downloaded in accordance with the previously described methods.[10] Participants used the same global positioning system device (S5; Catapult Sports, Melbourne, Australia) for each session to mitigate interunit measurement errors.[11] Distance covered per minute (in m/min) and percent of total distance completed above 14.4 km/h (percent high-speed running [%HSR]) were selected and used in subsequent statistical analyses, in part, to reduce potential issues of multicollinearity. Individual training session data were included in the analysis only if the athlete had completed a minimum of 80% of each prescribed training session.

Internal Training Load

The RPE was obtained 10 to 30 minutes following the completion of each training session using Borg CR-10 scale.[12] HR data were collected via chest strap HR monitors (T34; Polar Electro, Espoo, Finland). HR data were analyzed by quantifying the percent of total duration within specific "zones" (zone 3: 65%–74%, zone 4: 75%–84%, and zone 5: >85% HR_{max}).

Individual self-reported ratings of well-being were assessed via a psychometric questionnaire on a 10-point Likert scale with 1 representing "the worst I could possibly feel" and 10 representing "the best I could possibly feel" in accordance with the methods described previously.[13] Objective measures of heart-rate variability were assessed upon waking each morning by R-R series recording via photoplethysmography using a valid and reliable, commercially available smartphone application (HRV4Training).[14] Heart-rate variability data were subsequently analyzed for the root mean sum of the squared differences (rMSSD) between each successive heartbeat on recovery days +1 (24 h) and +2 (48 h) after training sessions, and compared with rolling baseline data obtained from a minimum of 4 readings in the prior 7 days.[15] rMSSD was chosen as the HR variable of interest due to the relationship with vagal activity[16] and greater reliability compared with other spectral indices.[17]

Environmental Monitoring

The training location at which data were collected was a coastal, subtropical region (28° S, 153° E) in Australia. The T_a (in °C) and RH (in percentage) were measured via a portable weather station (Kestrel 5000; Kestrel Instruments, Boothwyn, PA), whereas SR (in W/m^2) was recorded via pyranometer (MP-100; Apogee Instruments, Logan, UT) at 15-minute intervals during each field-based training session. The devices were mounted on a level tripod 1.5 m above the ground in the same location adjacent to the training field. After completion of each training session, data were downloaded to a custom Microsoft Excel spreadsheet (Microsoft Corp, Redmond, WA). To account for the varying duration of training sessions, environmental "exposure" was quantified by multiplying session duration by the mean of recorded T_a, RH, and SR with data expressed as session means.

Statistical Analyses

Training environment data during the experimental period were analyzed using one-way analysis of variance (SPSS Statistics, version 25; IBM Corp, Armonk, NY). Data are presented as mean (SD) with statistical significance set at $P < .05$.

Relationships between internal and external load and the training environment were analyzed using mixed-effects linear models via the Lme4 package in R Studio statistical computing software (version 1.1.442). Mixed-effect linear models were applied to training and recovery variables incorporating the individual as a random effect and heat stress variables as fixed effects using the equation:

$$y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + b_i + \varepsilon_i$$

where y_i is the value of the outcome variable of interest i, β_1 through β_3 are fixed-effect coefficients, x_{1i} through x_{3i} are fixed-effect variables for observation i, b_i is the random effects assumed to be multivariate normally distributed, and ε_i is the error for case i, which is also assumed to be multivariate normally distributed.[18] Normality assumptions were validated using residual and Q–Q plots, and the adequacy of the model structures was determined via residual plots and quantified using standard measures of intraclass correlations and coefficients of determination. All variables of interest in the model are shown as standardized regression coefficients (standardized beta [β]), allowing assessment of practical significance. Standardized regression coefficients for each variable were multiplied by the SD of the change in dependent variable to obtain the absolute change in the units of measurement.[19] Qualitative descriptors for reporting of standardized beta scores were adopted using a comparable approach to effect size statistics to interpret the magnitude of the association between heat stress, training, and recovery variables. We interpreted the β using threshold values of 0.2 as a small effect, 0.5 as a moderate effect, and 0.8 as a large effect and reported the 95% confidence intervals (95% CI). Multicollinearity between heat stress variables was established through Spearman correlation matrix analysis, and in

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the event that any variables had a multicollinearity $r \geq .8$, they were excluded from the model.[20,21]

Results

Environmental Conditions

Mean environmental conditions during the experimental period were 30.9°C (2.1°C) T_a (range: 26.7°C–34.4°C), 61% (6%) RH (range 52%–75%), and 718 (224) W/m^2 SR (range: 239–1001 W/m^2). Multicollinearity did not exist between heat stress variables. Wet-bulb globe temperature during the experimental period was 29°C (2.5°C; range: 24.4°C–32.9°C). There was no significant difference in variables of environmental conditions (Table 1) or training load (Table 2) between training sessions (n = 20) throughout the 8-week experimental period.

External Training Load

Increasing SR exposure was associated with a decrease in distance covered (−19.7 m/min; 95% CI, −21.4 to −18.0 m/min; $P < .001$), but there was no significant relationship between changes in RH and m/min at each training session (Figure 1A). In contrast, increasing T_a exposure was associated with an increase in distance covered (19.2 m/min; 95% CI, 18.0 to 21.4 m/min; $P < .001$). There were divergent effects on the change in %HSR completed during training sessions by each of the individual heat stress variables (Figure 1B). Specifically, an increase in SR exposure was associated with a large decrease in %HSR completed during training (−10.1%; 95% CI, −11.0% to −9.2%; $P < .001$). There was also a small effect of increasing RH that was associated with lower %HSR (−3.4%; CI, −4.2% to −2.5%; $P = .010$). However, increasing T_a exposure during training was related to a small increase in %HSR (3.5%; 95% CI, 2.7% to 4.4%; $P = .005$; Figure 1B).

Internal Training Load

Increasing T_a exposure during training sessions was associated with a concomitant increase in mean HR (4.8 bpm; 95% CI, 3.9 to 5.7 bpm; $P < .001$). However, there was no relationship between increases in SR or RH exposure and mean HR (Figure 2A). There were no significant relationships between the percent duration of training completed between 65% and 74% HR_{max} and any heat stress variables during the experimental period. Percent duration of training completed between 75% and 84% HR_{max} was associated with higher RH exposure (4.6%; 95% CI, 3.5% to 5.7%; $P = .002$), but there was no effect of either SR or T_a exposure on the percent duration of training completed between 75% and 84% HR_{max}. When RH exposure increased, there was an associated increase in percent duration of training being completed above 85% of HR_{max} (3.9%; 95% CI, 2.7% to 5.2%; $P < .001$) with a similar result evident with T_a exposure (4.3%; 95% CI, 3.1% to 5.6%; $P = .001$). In contrast, the percent duration of training sessions completed above 85% HR_{max} decreased with an associated increase in SR exposure (−2.9%; 95% CI, −4.1% to −1.7%; $P < .001$; Figure 2B).

Increasing RH exposure was associated with an increase in RPE (1.8 AU; 95% CI, 1.7 to 2.0 AU; $P < .001$). In addition, increased T_a exposure was also associated with an increase in RPE, but this was a small effect (0.3 AU; 95% CI, 0.1 to 0.4 AU; $P = .019$). There were no significant relationships between RPE and the level of SR exposure during training sessions.

Table 1 Environmental Conditions During the 8-Week Data-Collection Period

Heat-stress variable	Wks 1–8 average	Wks 1–8 median	Wks 1–8 range
Ambient temperature, °C	30.9 (2.1)	31.4	26.7–34.4
Relative humidity, %	61.7 (6.2)	61.2	52–75
Solar radiation, W/m^2	718 (224)	789	239–1001
Wind speed, kph	4.0 (1.3)	4.1	1.9–6.5
Wet-bulb-globe temperature, °C	29.0 (2.5)	29.6	24.4–32.9

Note: Data are represented as mean (SD) (n = 20 training sessions).

Table 2 Training Session Load Quantified via GPS, Electronic Heart-Rate Monitoring, and Participant RPE During the 8-Week Experimental Period for Players Preparing for Australian Football Competition (N = 45)

Training variable	Weeks 1–8, Mean (SD)	Median	Range
Session duration, min	70.8 (24.1)	60.4	26.7–97.7
Session distance covered, m	7323 (2853)	6417	2664–10,197
Average speed, m/min	102.7(21.1)	102.9	65.3–210.9
High-speed running, %	25.0 (10.6)	26.7	7.1–39.6
Mean HR, beats/min	155.7 (10.7)	154.6	140.4–163.3
Time >85% HR_{max}, %	16.9(15.1)	15.5	16.4–53.6
RPE, AU	6.4(1.9)	6.8	2.7–9.0
rMSSD +1, ms	114(31)	109	44–165
rMSSD +2, ms	127(35)	134	56–183

Abbreviations: GPS, global positioning system, HR, heart rate; rMSSD, root mean square of the successive differences; RPE, rating of perceived exertion.

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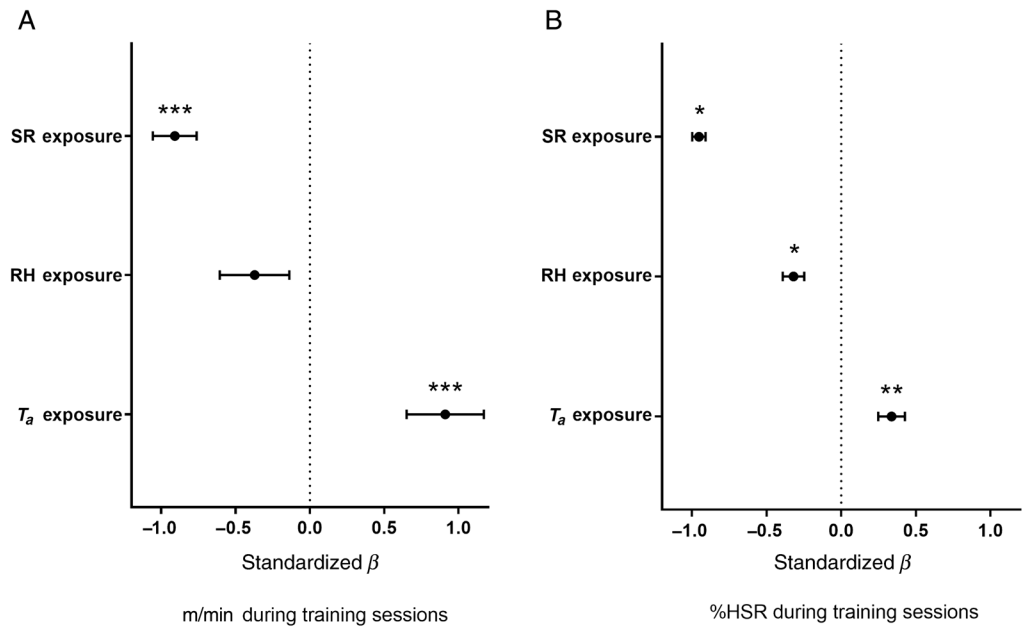

Figure 1 — Standardized coefficient relationships between (A) distance covered (in m/min) and heat-stress variables and (B) %HSR and heat-stress variables during an 8-week experimental period for players preparing for Australian Football competition (N = 45; 20 training sessions). SR indicates solar radiation; RH, relative humidity; T_a, ambient temperature; %HSR, percentage high-speed running. *$P < .05$. **$P < .01$. ***$P < .001$.

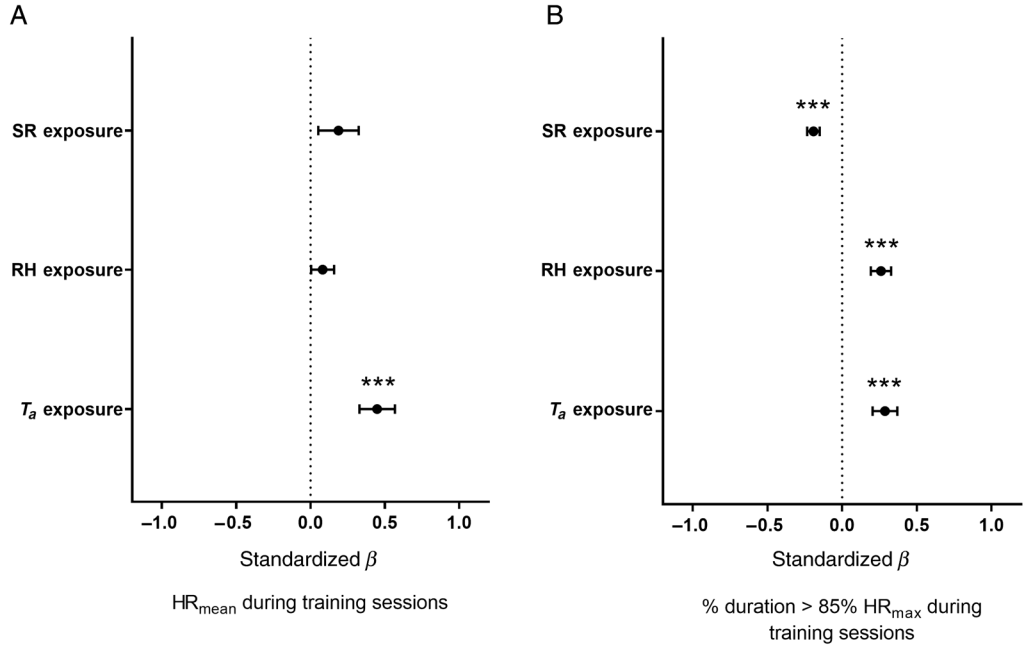

Figure 2 — Standardized coefficient relationship between (A) HR_{mean} and heat-stress variables and (B) percentage duration above 85% of maximal heart rate and heat-stress variables during an 8-week experimental period for players preparing for Australian Football competition (N = 45; 20 training sessions). HR_{mean} indicates mean heart rate; SR, solar radiation; RH, relative humidity; T_a, ambient temperature. ***$P < .001$.

Recovery

There was no effect of SR exposure on the BM change from presession to postsession during training, but there was an associated increase in BM loss with increasing RH (–215 g; 95% CI, –222 to –208 g; $P < .001$) and T_a (–160 g; 95% CI, –167 to –153 g; $P = .026$). No effect of any heat stress variable on self-reported "overall wellness" at 24 and 48 hours was evident. However, increasing RH exposure was associated with higher self-reported stress 24 hours posttraining (–0.11 AU; 95% CI, –0.17 to –0.05 AU; $P = .002$). Increasing RH exposure was not associated with self-reported recovery for the individual variables of "fatigue," "sleep quality," and "mood." Increasing T_a exposure was associated with a decrease in self-reported sleep quality 48 hours posttraining (–0.57 AU; 95% CI, –0.68 to –0.46 AU; $P = .03$), but no other self-reported recovery variable 48 hours posttraining was

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associated with heat stress variables. There was no effect of any heat stress variable on rMSSD 24 hours posttraining (Figure 3A). However, increasing SR exposure was associated with reduced rMSSD 48 hours posttraining (–16.9 ms; 95% CI, –24.6 to –9.3 ms; $P = .019$), but neither RH nor T_a exposure generated any significant effect on rMSSD 48 hours posttraining (Figure 3B).

Discussion

This study aimed to determine the individual effects of T_a, RH, and SR during physical preparation for professional team sport. Our data show the importance of considering the impact of the individual heat stress variables contributing to thermal load in isolation. Specifically, we show for the first time that SR is associated with profound effects on training in team-sport athletes related to reduced self-paced high-intensity work performed during the preparation phase of a competitive season. Our findings also show that RH is associated with reductions in the level of high-speed running during team-sport training and exerts the largest effect for increasing players' RPE. In contrast, increasing T_a was associated with higher work capacity during preseason training as evidenced by greater external training loads. Accordingly, the novel data from the present study indicate SR and RH each appear to have negative effects on the team-sport athlete, with associated reductions in intermittent high-intensity running capacity and the potential to slow recovery between training sessions.

Our data are in agreement with the limited number of previous studies investigating the effects of SR on exercise intensity or duration.[5,22] Otani et al[5] have previously shown time-to-exhaustion during prolonged endurance exercise is reduced even at moderate SR intensity (500 W/m^2) and that self-selected exercise intensity decreases when exposed to increasing levels of SR in hot environments.[22] We extend on these findings to show high SR exposure is closely related to impaired high-intensity work capacity during training for a professional team sport.

The associated decrease in high-intensity work performed with increased SR may be related to T_{sk}, which is heavily influenced by the external environment.[3] SR exposure has been shown to increase T_{sk} in a dose-response manner in thermoneutral[23,24] and hot[22,25] environments, while having little meaningful effect on core temperature.[5,24] The athletes in the current study routinely undertook training under SR intensities equivalent to those reported in previous studies showing increased T_{sk}. Consequently, we suggest that elevated T_{sk} with high SR exposure in the present study was likely a primary factor attenuating the capacity for prolonged high-intensity, intermittent exercise above that associated with high T_a alone. When T_{sk} increases, the core-to-skin temperature gradient narrows and promotes increases in skin blood flow, a decrease in stroke volume, and compromised cardiac output.[26] During self-paced team-sport training that includes repeated high-speed running in hot conditions, the thermoregulatory response limits exercise intensity to reduce metabolic heat production so that levels of compensable heat strain can be maintained.[27] Although we cannot ascribe cause and effect from our data, increased T_{sk}[28] and thermal perceptions of hot skin[29] have previously been associated with a decrease in intensity of aerobic exercise performance. Therefore, we propose that the team-sport athletes in the current study downregulated effort and intensity of work in response to higher T_{sk} with increasing SR exposure.

A second major finding of the present study was that RH was also associated with compromised external work and higher internal stress. It is well established that when RH is low, evaporative cooling is an efficient cooling mechanism and that increasing humidity limits evaporation, as sweat secreted to the skin surface is not easily dissipated to the external environment.[4] Moreover, in hot conditions, the body will also gain heat from the environment through radiation and conduction.[1] In the present study, increasing RH exposure was associated with a small-to-moderate effect on work performed during training but had the largest effect on an individual's perception of exertion. When considering the hot and

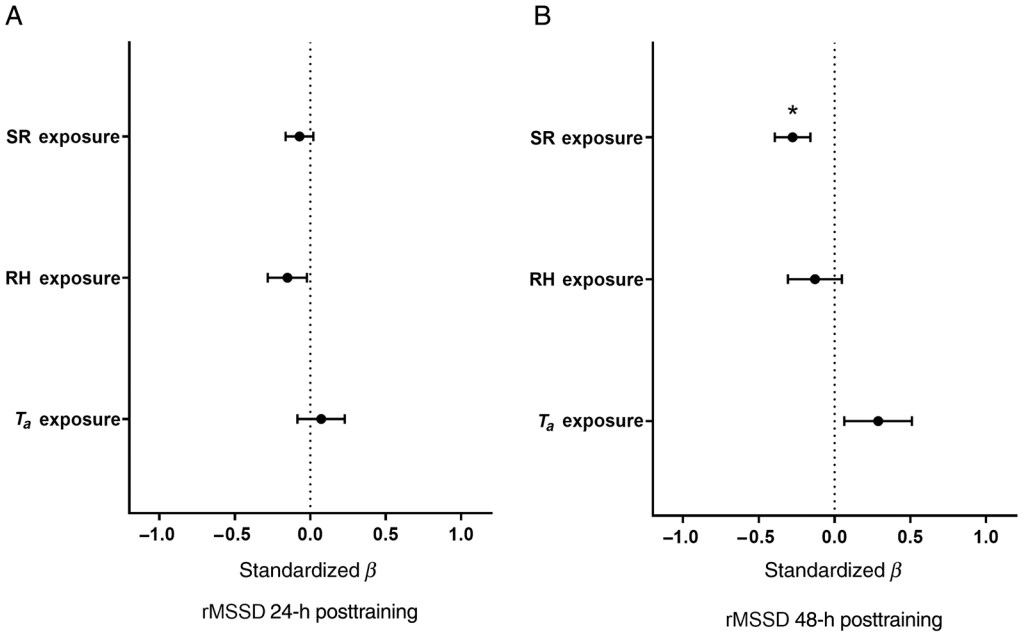

Figure 3 — Standardized coefficient relationship between heart-rate variability (rMSSD) and heat-stress variables following (A) 24- and (B) 48-hour recovery from training sessions during an 8-week experimental period for players preparing for Australian Football competition (N = 45; 20 training sessions). SR indicates solar radiation; RH, relative humidity; T_a, ambient temperature; rMSSD, root mean sum of the squared differences. *$P < .05$.

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humid typical training environment of the present study, the effect of RH on perceived effort and HR may have been expected. However, this effect did not appear to decrease external work output to the same extent as SR, indicating there may be incongruence between external and internal load parameters during preparation for team sport in hot environments with potential implications for training load monitoring of athletes in the heat.

In contrast to the effects of SR and RH, higher T_a was associated with increased work performed and internal load during training sessions. Importantly, an operational construct of the statistical model employed is that the effects of an increasing individual heat stress variable are determined within a paradigm where the other heat variables in the model remain constant. Under conditions of moderate-to-high T_a (~30°C–35°C), it has been purported that the increased muscle temperature with passive heat exposure may promote improved performance capacity for repeated, high-intensity efforts similar to those undertaken in team-sport training.[30] This phenomenon may, at least in part, explain the association between increasing T_a and running performance in the present study. While the associated increase in HR with RH likely reflects the effects of heat strain, the association between elevated HR and increased T_a may be related to the greater work performed with increasing T_a. Moreover, high levels of aerobic fitness are closely associated with effective heat mitigation, and together with heat acclimation may represent the primary strategies for enhancing exercise capacity in hot conditions.[31] Participants in our study were heat-acclimatized professional team-sport athletes with a high level of aerobic fitness and a well-developed capability to tolerate stressful training environments. As such, it seems reasonable to suggest that increased muscle temperature could be achieved in trained athletes without or in spite of an excessive rise in core temperature, a response that may also be a prerequisite for improved performance of repeated high-intensity efforts in the heat.[30] However, it is possible that the magnitude of heat stress from increasing T_a exposure in the present study was insufficient to elicit detrimental effects more commonly associated with heat stress protocols in laboratory settings. Moreover, it is unclear if similar responses to increasing T_a exposure would be observed in different athlete cohorts, geographical locations, or while undertaking different training protocols.

Intuitively, the physiological strain of undertaking high-intensity team-sport training in hot environments would elicit a significant effect on psychometric well-being and parasympathetic activity that is reflected in rMSSD.[32] Indeed, high-intensity exercise (>50% VO_2peak) has been shown to progressively decrease parasympathetic tone[33] with the potential to increase subjective feelings of fatigue and decrease sleep quality[33] during recovery from exercise in hot conditions. The lack of association between heat stress variables and most subjective and objective recovery measures after 24-hour recovery was unexpected, although there was a modest association between RH and self-reported stress the day after training. However, a recently published study investigating the recovery time course in Australian Football has shown that the stress response appears to reach a peak after ~40-hour recovery from competition.[34] Our data show a small associated decrease in rMSSD 48 hours after training with increasing SR exposure, an effect that was not apparent in response to increasing T_a or RH. In addition, T_a was associated with a decrease in self-reported sleep quality 48 hours posttraining with no other subjective recovery measures associated with heat stress variables. It could be that quantification of acute heart-rate variability has limited capacity to detect a meaningful change in the current study and chronic responses analyzed over prolonged periods may be more informative.[35] Our varied findings may also simply reflect the complexity of interactions in subjective and objective measures of recovery,[36] and the most representative sample to detect important relationships is also unknown. Further research is required to determine the relationships between team-sport athlete training load, measures of recovery and heat stress variables, and the effect on athlete well-being during preparation for competition.

Practical Applications

Measuring environmental parameters individually may be more sensitive when assessing the magnitude of heat stress during team-sport training and recovery. Moreover, heat management strategies should not be limited to competition, but it should include the physical preparation period if maximizing the quality and quantity of work completed within team-sport training is desired.

Conclusions

In conclusion, our data are the first to show the effects of individual heat stress variables on team-sport training and the associated acute negative effects on physical performance and recovery parameters with exposure to increasing SR and RH. Indeed, we have demonstrated these effects in heat-acclimatized athletes indicating that heat acclimation alone may be unable to ameliorate the reduced acute responses associated with team-sport training undertaken in hot conditions. Given the reality of a warming climate and the increased prevalence of environmental extremes in many geographical locations, mitigating the effects of heat for team-sport athletes will be increasingly important.

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References

1. Sawka MN, Leon LR, Montain SJ, Sonna LA. Integrated physiological mechanisms of exercise performance, adaptation, and maladaptation to heat stress. *Compr Physiol*. 2011;1(4):1883–1928. PubMed ID: 23733692 doi:10.1002/cphy.c100082
2. Cheuvront SN, Kenefick RW, Montain SJ, Sawka MN. Mechanisms of aerobic performance impairment with heat stress and dehydration. *J Appl Physiol*. 2010;109(6):1989–1995. PubMed ID: 20689090 doi:10.1152/japplphysiol.00367.2010
3. Galloway SD, Maughan RJ. Effects of ambient temperature on the capacity to perform prolonged cycle exercise in man. *Med Sci Sports Exerc*. 1997;29(9):1240–1249. PubMed ID: 9309637 doi:10.1097/00005768-199709000-00018
4. Maughan RJ, Otani H, Watson P. Influence of relative humidity on prolonged exercise capacity in a warm environment. *Eur J Appl Physiol*. 2012;112(6):2313–2321. PubMed ID: 22012542 doi:10.1007/s00421-011-2206-7
5. Otani H, Kaya M, Tamaki A, Watson P, Maughan RJ. Effects of solar radiation on endurance exercise capacity in a hot environment. *Eur J Appl Physiol*. 2016;116(4):769–779. PubMed ID: 26842928 doi:10.1007/s00421-016-3335-9
6. Slattery K, Coutts A. The application of heat stress to team sports: football/soccer, Australian football and rugby. In: Périard JD,

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Racinais S, eds. *Heat Stress in Sport and Exercise: Thermophysiology of Health and Performance*. Berlin, Germany: Springer; 2019: 181–202.
7. Aughey RJ, Goodman CA, McKenna MJ. Greater chance of high core temperatures with modified pacing strategy during team sport in the heat. *J Sci Med Sport*. 2014;17(1):113–118. PubMed ID: 23689104 doi:10.1016/j.jsams.2013.02.013
8. Pryor JL, Johnson EC, Roberts WO, Pryor RR. Application of evidence-based recommendations for heat acclimation: individual and team sport perspectives. *Temperature*. 2019;6(1):37–49. doi:10.1080/23328940.2018.1516537
9. Schmit C, Duffield R, Hausswirth C, Brisswalter J, Le Meur Y. Optimizing heat acclimation for endurance athletes: high- versus low-intensity training. *Int J Sports Physiol Perform*. 2018;13(6):816–823. PubMed ID: 28872380 doi:10.1123/ijspp.2017-0007
10. Bartlett JD, O'Connor F, Pitchford N, Torres-Ronda L, Robertson SJ. Relationships between internal and external training load in team-sport athletes: evidence for an individualized approach. *Int J Sports Physiol Perform*. 2017;12(2):230–234. PubMed ID: 27194668 doi:10.1123/ijspp.2015-0791
11. Coutts AJ, Duffield R. Validity and reliability of GPS devices for measuring movement demands of team sports. *J Sci Med Sport*. 2010;13(1):133–135. PubMed ID: 19054711 doi:10.1016/j.jsams.2008.09.015
12. Borg GA. Psychophysical bases of perceived exertion. *Med Sci Sports Exerc*. 1982;14(5):377–381. PubMed ID: 7154893 doi:10.1249/00005768-198205000-00012
13. Gallo TF, Cormack SJ, Gabbett TJ, Lorenzen CH. Self-reported wellness profiles of professional Australian football players during the competition phase of the season. *J Strength Cond Res*. 2017;31(2): 495–502. PubMed ID: 27243912 doi:10.1519/JSC.0000000000001515
14. Plews DJ, Scott B, Altini M, Wood M, Kilding AE, Laursen PB. Comparison of heart-rate-variability recording with smartphone photoplethysmography, polar H7 chest strap, and electrocardiography. *Int J Sports Physiol Perform*. 2017;12(10):1324–1328. PubMed ID: 28290720 doi:10.1123/ijspp.2016-0668
15. Plews DJ, Laursen PB, Le Meur Y, Hausswirth C, Kilding AE, Buchheit M. Monitoring training with heart ratevariability: how much compliance is needed for valid assessment? *Int J Sports Physiol Perform*. 2014;9(5):783–790. PubMed ID: 24334285 doi:10.1123/ijspp.2013-0455
16. Malik M, Bigger JT, Camm AJ, et al. Heart rate variability standards of measurement, physiological interpretation, and clinical use. *Eur Heart J*. 1996;17(3):354–381. doi:10.1093/oxfordjournals.eurheartj.a014868
17. Al Haddad H, Laursen PB, Chollet D, Ahmaidi S, Buchheit M. Reliability of resting and postexercise heart rate measures. *Int J Sports Med*. 2011;32(8):598–605. PubMed ID: 21574126 doi:10.1055/s-0031-1275356
18. Bates D, Mächler M, Bolker B, Walker S. Fitting linear mixed-effects models using lme4. *J Stat Software*. 2015;67(1):48. doi:10.18637/jss.v067.i01
19. Nieminen P, Lehtiniemi H, Vähäkangas K, Huusko A, Rautio A. Standardised regression coefficient as an effect size index in summarising findings in epidemiological studies. *Epidemiol Biostat Public Health*. 2013;10(4):e8854-1–e8854-15.
20. Robertson S, Back N, Bartlett JD. Explaining match outcome in elite Australian rules football using team performance indicators. *J Sports Sci*. 2016;34(7):637–644. PubMed ID: 26176890 doi:10.1080/02640414.2015.1066026
21. Robertson S, Gupta R, McIntosh S. A method to assess the influence of individual player performance distribution on match outcome in team sports. *J Sports Sci*. 2016;34(19):1893–1900. PubMed ID: 26853070 doi:10.1080/02640414.2016.1142106
22. Otani H, Kaya M, Tamaki A, Goto H, Maughan RJ. Exposure to high solar radiation reduces self-regulated exercise intensity in the heat outdoors. *Physiol Behav*. 2019;199:191–199. PubMed ID: 30471385 doi:10.1016/j.physbeh.2018.11.029
23. Hodder SG, Parsons K. The effects of solar radiation on thermal comfort. *Int J Biometeorol*. 2007;51(3):233–250. PubMed ID: 17009012 doi:10.1007/s00484-006-0050-y
24. Nielsen B, Kassow K, Aschengreen FE. Heat balance during exercise in the sun. *Eur J Appl Physiol Occup Physiol*. 1988;58(1–2):189–196. doi:10.1007/BF00636625
25. Otani H, Goto T, Goto H, Shirato M. Time-of-day effects of exposure to solar radiation on thermoregulation during outdoor exercise in the heat. *Chronobiol Int*. 2017;34(9):1224–1238. PubMed ID: 28910548 doi:10.1080/07420528.2017.1358735
26. Ely BR, Ely MR, Cheuvront SN, Kenefick RW, Degroot DW, Montain SJ. Evidence against a 40 degrees C core temperature threshold for fatigue in humans. *J Appl Physiol*. 2009;107(5):1519–1525. PubMed ID: 19713430 doi:10.1152/japplphysiol.00577.2009
27. Schlader ZJ, Stannard SR, Mundel T. Exercise and heat stress: performance, fatigue and exhaustion–a hot topic. *Br J Sports Med*. 2011;45(1): 3–5. PubMed ID: 19846428 doi:10.1136/bjsm.2009.063024
28. Schlader ZJ, Simmons SE, Stannard SR, Mundel T. Skin temperature as a thermal controller of exercise intensity. *Eur J Appl Physiol*. 2011;111(8):1631–1639. PubMed ID: 21197543 doi:10.1007/s00421-010-1791-1
29. Schlader ZJ, Simmons SE, Stannard SR, Mundel T. The independent roles of temperature and thermal perception in the control of human thermoregulatory behavior. *Physiol Behav*. 2011;103(2):217–224. PubMed ID: 21315099 doi:10.1016/j.physbeh.2011.02.002
30. Girard O, Brocherie F, Bishop DJ. Sprint performance under heat stress: a review. *Scand J Med Sci Sports*. 2015;25(Suppl 1):79–89. doi:10.1111/sms.2015.25.issue-s1
31. Alhadad SB, Tan PMS, Lee JKW. Efficacy of heat mitigation strategies on core temperature and endurance exercise: a meta-analysis. *Front Physiol*. 2019;10:71. PubMed ID: 30842739 doi:10.3389/fphys.2019.00071
32. Shaffer F, Ginsberg JP. An overview of heart rate variability metrics and norms. *Front Public Health*. 2017;5:258–258. PubMed ID: 29034226 doi:10.3389/fpubh.2017.00258
33. Buchheit M, Cholley Y, Lambert P. Psychometric and physiological responses to a preseason competitive camp in the heat with a 6-hour time difference in elite soccer players. *Int J Sports Physiol Perform*. 2016;11(2):176–181. PubMed ID: 26182437 doi:10.1123/ijspp.2015-0135
34. Gastin PB, Hunkin SL, Fahrner B, Robertson S. Deceleration, acceleration, and impacts are strong contributors to muscle damage in professional Australian football. *J Strength Cond Res*. 2019;33(12), 3374–3383. PubMed ID: 30694964 doi:10.1519/JSC.0000000000003023
35. Plews DJ, Laursen PB, Stanley J, Kilding AE, Buchheit M. Training adaptation and heart rate variability in elite endurance athletes: opening the door to effective monitoring. *Sports Med*. 2013;43(9):773–781. PubMed ID: 23852425 doi:10.1007/s40279-013-0071-8
36. Saw AE, Main LC, Gastin PB. Monitoring the athlete training response: subjective self-reported measures trump commonly used objective measures: a systematic review. *Br J Sports Med*. 2016;50(5): 281–291. PubMed ID: 26423706 doi:10.1136/bjsports-2015-094758

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